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RESEARCH ARTICLE

Reimagining Lunar Dust: A Novel Reinforcement for Fiber-Reinforced Polymer Matrix Composite Materials

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ABSTRACT

The prohibitive cost of transporting construction materials to space underscores the need for *in-situ* resource utilization strategies (ISRU). Lunar regolith dust represents an abundant and sustainable reinforcement for advanced composites. Here, we investigate the use of Black Point-1 (BP-1) lunar soil simulant as a novel filler in fiber-reinforced polymer composites. Composites containing 1–10 wt% BP-1 were systematically evaluated through tensile testing, interlaminar shear strength (ILSS) measurements, fracture toughness characterization, and low-velocity impact experiments. At optimal loadings, BP-1 increases ILSS by 30%–40% and impact resistance by 15% while preserving the intrinsic properties of the polymer matrix. Micro-toughening mechanisms, including fiber bridging and crack deflection promoted by particle–matrix interactions, effectively suppress interlaminar damage initiation and propagation. Compared to conventional fillers, BP-1 delivers superior reinforcement efficiency on a weight-normalized basis, attributed to its angular morphology and high surface area. These results demonstrate that lunar dust can serve as an effective reinforcement for fiber-reinforced composites, offering a resource-efficient pathway toward durable materials for extraterrestrial and terrestrial infrastructure.

1 | Introduction

The renewed drive for lunar exploration, coupled with the ambition to establish a sustained human presence on the Moon, has created an urgent need for advanced materials capable of enduring the Moon's extreme thermal, mechanical, and environmental conditions. Beyond enabling long-term habitation, such materials are vital for supporting deep-space missions and reducing reliance on costly Earth-based supply chains. Lunar regolith, one of the most abundant resources on the Moon, is a fine, angular particulate rich in oxides, silicates, and other minerals. While its abrasive nature and potential health risks present challenges, its chemical composition, mechanical hardness, and on-site availability make it an attractive candidate for *in-situ* resource utilization (ISRU). Leveraging lunar regolith in structural and manufacturing applications offers a pathway toward sustainable, cost-effective solutions for construction on the Moon, aligning with the broader vision of resilient and self-sufficient space infrastructure [1–6].

Historically, lunar regolith has been investigated for applications such as sintered blocks, radiation shielding, and lunar concrete [7–10]. More recently, extensive efforts have focused on regolith-based construction materials, including alkali-activated geopolymers, regolith concretes, and regolith bag or sintered structural systems. These material concepts primarily target compressive load-bearing functions, radiation protection, and habitat-scale infrastructure, leveraging high regolith volume fractions to minimize reliance on Earth-supplied materials. Promising mechanical performance and constructability under lunar-relevant conditions have been reported for these approaches [11–14]. However, its potential as a reinforcement material in fiber-reinforced polymer matrix composites (PMCs) remains underexplored. Fiber-reinforced PMC architectures address a distinct design space in which low density, tensile strength, fracture toughness, and impact resistance are critical. Incorporating lunar regolith as a particulate reinforcement in PMCs offers a pathway to enhance damage tolerance and interfacial performance while maintaining lightweight structural efficiency. Moreover, regolith-reinforced PMCs are compatible with advanced manufacturing routes, including additive and hybrid fabrication techniques, enabling on-demand production of structural components. Transforming lunar dust from a problematic contaminant into a value-added reinforcement thus represents a key step toward sustainable, resilient materials for long-term extraterrestrial exploration and infrastructure [6].

The use of particulate fillers to enhance PMCs has a long-standing history in engineering applications, serving as a cornerstone for the development of high-performance composite systems. These fillers are introduced into polymer matrices to tailor their mechanical, thermal, and fracture properties, thereby meeting the demands of various industrial applications. Conventional fillers, such as carbon nanofibers [15–25], graphene [26–33], and other ceramic particles [22, 31, 34–42] such as alumina, silica, and silicon carbide, have been extensively studied due to their exceptional ability to improve the overall performance of fiber-reinforced polymer composites. Carbon nanofibers and graphene, for instance, are renowned for their remarkable strength, stiffness, and electrical conductivity, enabling their use in advanced aerospace and electronic applications [17, 24, 29]. Their nanoscale structure provides a high surface area, promoting superior fiber/matrix load transfer and interfacial adhesion at both the lamina and laminate levels. Alumina and silica, on the other hand, are widely recognized for their thermal stability and resistance to wear [35], which make them suitable for high-temperature and high-stress environments, such as automotive components and industrial machinery. These fillers enhance the strength and stiffness of the matrix material while simultaneously improving its damage resistance and ability to withstand harsh operational conditions [34].

Recent advances in enhancing PMCs have increasingly focused on optimizing particulate fillers through morphology control, surface functionalization, and hybridization strategies [41, 43–51]. Such modifications improve filler dispersion and interfacial adhesion, enabling mechanisms such as crack deflection, fiber bridging, and stress redistribution that collectively enhance toughness, stiffness, and durability. Alongside these synthetic approaches, natural and mineral-based fillers, including volcanic ash, basalt fibers, and other mineral particulates, have attracted interest as sustainable alternatives [38, 52–55]. These materials offer comparable mechanical reinforcement with reduced processing demands, lower carbon footprint, and greater cost efficiency. Basalt fibers, for example, combine a high strength-to-weight ratio with corrosion resistance, making them a lightweight and durable option for aerospace and marine composites. Their growing use underscores a broader shift toward sustainable, resource-efficient composite technologies.

Despite these advances, the reinforcement of composites with extraterrestrial materials remains largely unexplored. While lunar regolith has been preliminarily examined in bulk applications such as cementitious binders and structural concrete analogs [7–10], its potential as a functional filler in fiber-reinforced polymer composites has not been systematically investigated. Exploring such unconventional reinforcements offers not only a pathway to sustainable space infrastructure but also opportunities to design next-generation composites that leverage abundant, underutilized resources.

Given its abundance and mineral-rich composition, lunar regolith represents a unique a particulate filler for composite materials specifically tailored to extraterrestrial applications. Beyond its *in-situ* availability under ISRU strategies, regolith offers the potential to enhance tensile strength, stiffness, wear resistance, and environmental durability in PMCs. Unlike conventional fillers, regolith possesses a complex and highly variable composition, which is dominated by silicates, oxides, and metallic phases, along with an irregular, sharp-edged morphology that may promote mechanical interlocking and improve interfacial load transfer. Its high thermal stability and inherent radiation resistance further position regolith-based composites as promising candidates for use under the extreme temperature fluctuations and high-energy radiation of lunar environments.

Nonetheless, challenges remain. The abrasive character of regolith particles can complicate processing, accelerate tool wear, and increase fiber abrasion, necessitating tailored surface treatments or dispersion strategies to optimize compatibility with polymer matrices. Preliminary demonstrations of extraterrestrial materials in other contexts, such as sintered regolith tiles and regolith-based concretes, underscore the feasibility of developing structural materials directly from lunar soil. Extending this concept to fiber-reinforced polymer composites could enable lightweight, durable components for structural panels, protective covers, and rover or habitat elements, addressing immediate mission needs while laying a foundation for broader space exploration goals, including Mars missions.

To experimentally explore this potential, this study investigates the potential of lunar regolith dust, specifically Black Point-1 (BP-1) lunar soil simulant, as a reinforcement material for glass fiber-reinforced polymer composites (GFRP). BP-1 is a widely used terrestrial simulant derived from basaltic volcanic deposits and is designed to replicate the particle size distribution, morphology, and mineralogical composition of lunar regolith. The particles exhibit a heterogeneous, predominantly angular morphology with an average size of approximately 6–7 μm (Figure 1a,b) and contain major oxide-forming elements characteristic of lunar soil, as confirmed by scanning electron microscopy (SEM) and EDS analyses (Figure 1c).

BP-1-reinforced GFRP laminates were fabricated using a conventional hand layup and vacuum-assisted consolidation process, producing well-consolidated composites with uniform particle dispersion and robust fiber–matrix bonding (Figure 1d,e). Mechanical and fracture performance was evaluated through tensile testing, single-edge notched bending (SENB), short-beam shear (SBS) tests, and low-velocity impact experiments. Together, these measurements provide a comprehensive assessment of stiffness, strength, fracture resistance, interlaminar shear behavior, and impact damage tolerance in regolith-reinforced composite systems. By demonstrating how modest regolith additions can enhance damage tolerance and interfacial performance in lightweight composites, this work advances ISRU-oriented materials strategies and supports the development of durable, resource-efficient structural materials for future lunar exploration and habitation.

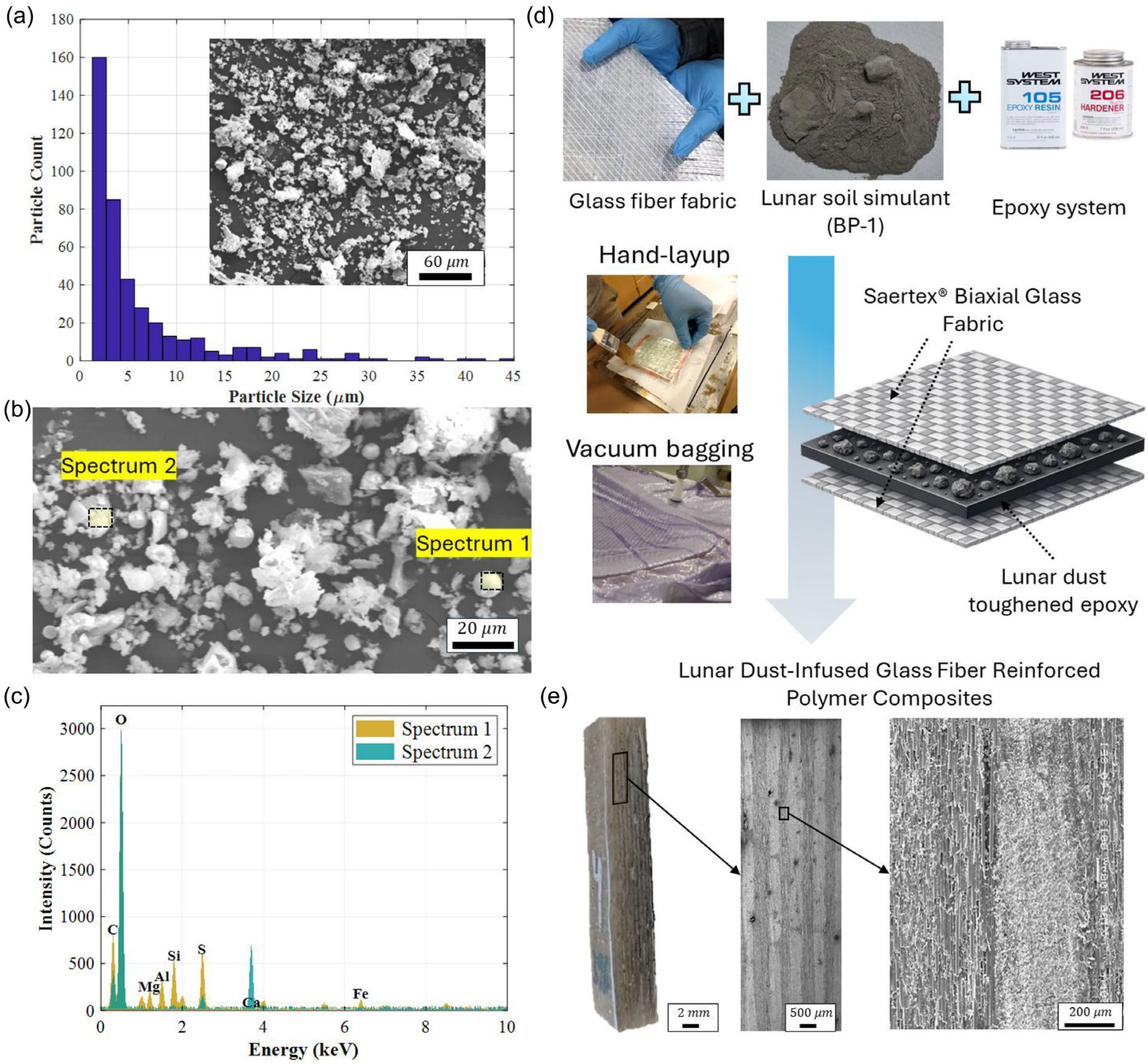

FIGURE 1 | (a) Particle size distribution of the BP-1 lunar regolith simulant, exhibiting an average particle size of approximately 6–7 μm; the inset shows a representative scanned area used for analysis. (b) High-magnification SEM images of two representative particles within the BP-1 simulant, highlighting particle morphology and angularity. (c) Energy-dispersive X-ray spectroscopy (EDS) spectra obtained from Spectrum 1 and Spectrum 2 regions, confirming elemental composition. (d) Materials and fabrication workflow for producing lunar dust-toughened glass fiber/epoxy composites, including glass fiber fabric, BP-1 simulant, epoxy resin, layup procedure, and schematic of the final laminate architecture. (e) Macro- and microstructural cross-sectional examination of the fabricated composite at multiple magnifications; white particles correspond to lunar dust particles dispersed within the epoxy matrix.

2 | Results and Discussion

2.1 | Mechanical and Fracture Properties of Lunar Dust-Reinforced Epoxy

Understanding the mechanisms by which reinforcement particles interact with the polymer matrix is crucial when studying particle-reinforced polymer composites. Matrix–filler interactions are governed by particle size, morphology, surface chemistry, and dispersion within the polymer matrix, which collectively determine the quality of matrix–particle adhesion and the efficiency of load transfer. Effective interfacial adhesion is critical for enhancing mechanical performance, whereas poor dispersion or inadequate bonding can introduce local stress concentrations and degrade composite properties. In addition to influencing stiffness and strength, particulate reinforcement alters fracture behavior by affecting crack initiation and propagation. Mechanisms such as crack deflection, crack pinning, and the formation of tortuous fracture paths promote energy dissipation and contribute to improved toughness and damage tolerance.

Given these complexities, the first phase of this study systematically assesses the influence of BP-1 lunar dust simulant particles on the mechanical and fracture properties of epoxy. Tensile tests were used to characterize changes in stiffness, strength, and ductility, while SENB tests probed fracture toughness and crack resistance across a range of BP-1 concentrations (1–8 wt%).

The tensile response (Figure 2a) reveals a clear stiffening effect resulting from BP-1 addition. Neat epoxy exhibits higher ductility, whereas particle reinforcement leads to a reduction in strain

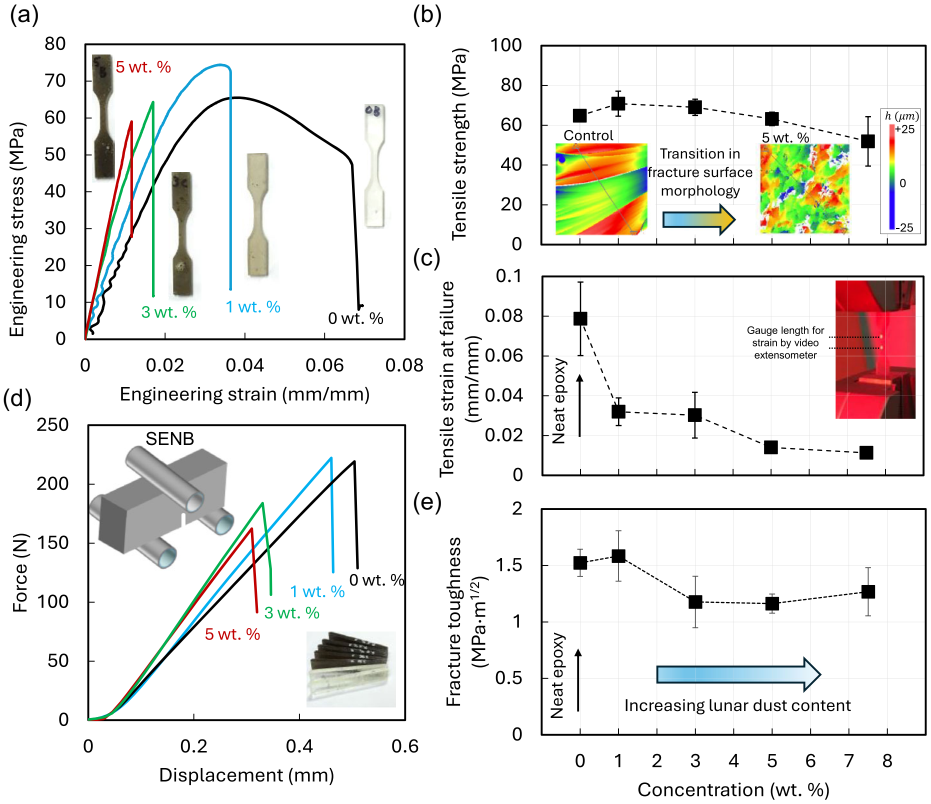

FIGURE 2 | (a) Representative engineering tensile stress–strain curves for neat epoxy and BP-1 particle-reinforced epoxy at 1, 3, and 5 wt%; inset shows three-dimensional (3D) topography of fracture surfaces measured by noncontact surface profilometry for neat epoxy and 5 wt% samples. (b) Ultimate tensile strength as a function of BP-1 concentration. (c) Tensile strain at failure as a function of BP-1 concentration; inset indicates the gauge length used for strain measurement with a video extensometer. (d) Representative force–displacement curves from SENB tests for neat epoxy and BP-1-reinforced epoxy at 1, 3, and 5 wt%. (e) Fracture toughness of epoxy as a function of BP-1 concentration derived from SENB tests.

at failure accompanied by a modest increase in elastic stiffness of approximately 15%. The measured stiffness values showed minimal specimen-to-specimen variability across all compositions, with variability remaining within a few percent, indicating a robust and repeatable stiffening response. This trend is reflected in the ultimate tensile strength results (Figure 2b), where low filler loadings lead to a modest strength increase due to improved load transfer at the particle–matrix interface. At higher particle concentrations, however, tensile strength plateaus or slightly declines, consistent with the onset of particle agglomeration and a reduction in reinforcement efficiency. Corresponding fracture surface topography (inset, Figure 2b) shows a transition from relatively smooth, stepped fracture in neat epoxy to more tortuous, particle-mediated morphologies in reinforced samples. Strain-at-failure results (Figure 2c) confirm the ductility–stiffness trade-off, highlighting the perceived macroscopic embrittling effect of excessive particle addition.

Fracture behavior is also influenced by particle addition. Force–displacement curves from SENB tests (Figure 2d) demonstrate increased initial stiffness in reinforced samples but slightly lower peak loads relative to neat epoxy. The initial nonlinearity observed in these curves (retained during analysis) is attributed to seating effects arising from contact between the specimens and the loading/support pins. The calculated fracture toughness (Figure 2e) indicates that moderate filler content enhances resistance to crack growth via mechanisms such as crack deflection and local energy dissipation. At higher particle concentrations, however, agglomerates act as stress concentrators, reducing peak load-carrying capacity and overall fracture resistance.

These results show that BP-1 particles significantly alter the balance of mechanical and fracture properties in epoxy composites. The most favorable overall balance of properties is observed at modest filler loadings (~1 wt%), where stiffness and strength are maintained at levels comparable to neat epoxy while fracture toughness remains stable within experimental uncertainty. These results demonstrate that BP-1 particles can effectively tailor the stiffness and fracture behavior of epoxy systems, but only at low concentrations where dispersion is uniform. Beyond this threshold, agglomeration offsets the benefits, highlighting the importance of optimizing filler content to preserve matrix-dominated properties.

2.2 | Interlaminar Shear Strength of Lunar Dust-Reinforced Composites

In fiber-reinforced composites, interlaminar shear strength (ILSS) is a key parameter governing mechanical performance. ILSS reflects the ability of laminates to resist shear stresses at the fiber–matrix interface, which directly controls load transfer and damage initiation. Particle reinforcement can play an important role in both strengthening and toughening these interfaces, thereby improving overall composite reliability. This section

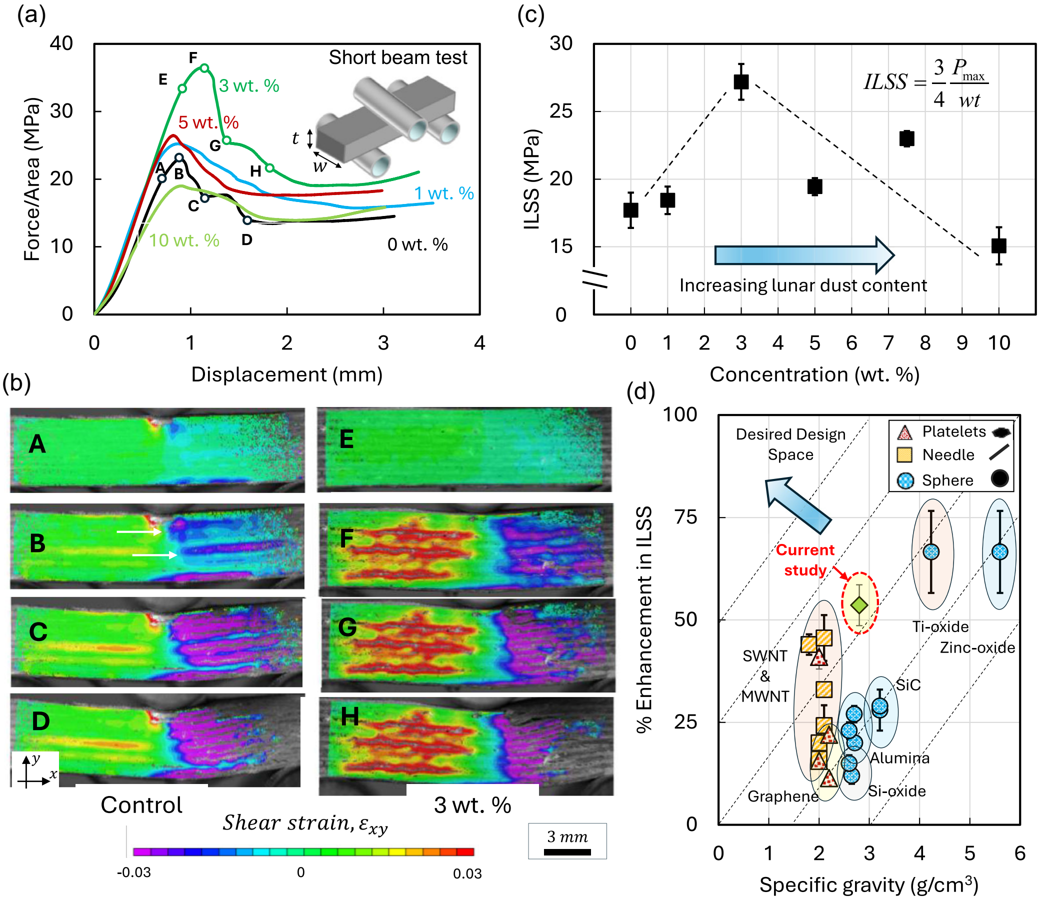

FIGURE 3 | (a) Normalized force (Force/cross-sectional Area) versus displacement curves obtained from short beam bending tests on glass fiber-reinforced composites with neat epoxy and BP-1 particle-reinforced epoxy at concentrations of 1, 3, 5, and 10 wt%. (b) In-plane shear strain fields measured via digital image correlation (DIC) on the side surfaces of control (neat epoxy) and 3 wt% BP-1 samples at various deformation stages, as marked in panel (a). (c) ILSS of the composites as a function of BP-1 lunar dust simulant concentration, demonstrating a significant enhancement with increasing filler content. (d) A comparative chart showing the percentage improvement in ILSS (relative to composites with meat epoxy matrices) achieved using BP-1 particles relative to common reinforcements such as spherical, platelet, and needle-shaped particles, as a function of filler-specific gravity, highlighting the superior performance of BP-1 particles compared to conventional fillers reported in the literature [14–20, 26–30, 34–39]. Data are normalized with respect to the neat epoxy baseline in each study; symbols denote mean relative changes, and error bars represent the range of reported values across different filler contents.

investigates the effect of BP-1 lunar dust simulant as a particulate reinforcement on the interlaminar shear behavior of GFRP laminates. SBS tests were employed to characterize ILSS and assess how BP-1 contributes to shear strength and deformation resistance in the composite system.

Figure 3 highlights the influence of BP-1 lunar dust particles on the ILSS and shear deformation behavior of GFRP laminates. Panel (a) presents normalized force–displacement curves from SBS tests for composites fabricated with neat epoxy and BP-1-reinforced epoxy at 1–10 wt%. Normalization by cross-sectional area accounts for minor sample thickness variations, enabling direct comparison. Additions of 1–5 wt% BP-1 increase the load-bearing capacity, as reflected in higher peak forces by up to 65% at 3% filler content, while the extended postpeak plateau increases by up to 43% relative to the neat epoxy matrix reference. These increases indicate improved "shear" strength at comparable elastic stiffness across the investigated filler concentration range, implying improved intrinsic resistance to interfacial shear damage. These results suggest that moderate BP-1 incorporation (around 3%) strengthens the epoxy matrix and enhances stress transfer across interlaminar regions, whereas excessive particle loadings reduce these gains due to particle agglomeration, leading to macroscopic embrittlement.

Panel (b) examines shear deformation fields obtained via DIC. Relative to neat epoxy, the 3 wt% BP-1 composite exhibits a more spatially distributed shear strain field across the beam cross section and along both positive and negative shear spans of the beam. This distributed deformation reflects reduced strain localization and delayed onset of interlaminar damage, indicative of improved load transfer within the laminate. Rather than concentrating shear deformation near a single delamination front, the BP-1–reinforced composite accommodates shear through a broader deformation zone. This behavior is attributed to effective particle dispersion within the matrix, which enhances matrix stiffness and interfacial resistance, mitigates local stress concentrations, and promotes stable shear deformation prior to failure. Panel (c) quantifies the ILSS as a function of BP-1 content. The shear strength increases substantially with particle addition, peaking at 3–7.5 wt%, followed by a plateau or slight decline at 10 wt%. This trend underlines the importance of optimizing filler concentration to balance reinforcement benefits against the adverse effects of particle clustering, as shown in Figure 2 for higher particle concentration.

Panel (d) expands the comparison by evaluating ILSS enhancements from BP-1 particles against those achieved with common fillers such as spherical, platelet, and needle-shaped particles, normalized by specific gravity. The data were compiled from multiple published studies and normalized against the neat epoxy baseline of each reference composite system to highlight relative property enhancements and enable consistent cross-study comparison. For each system, the plotted values represent the mean relative change, while the error bars indicate the range of reported property variations associated with different filler contents. A general trend is observed in which ILSS improvement increases with the specific gravity of the filler. Low-density carbon-based fillers including single- and multi-walled carbon nanotubes and graphene derivatives exhibit substantial enhancements in the range of 10%–50%, owing to their exceptionally low density [15–20]. Slightly heavier particles, such as silica, alumina, and silicon carbide, provide more moderate improvements of approximately 15%–25% [26–30]. Higher-density fillers, such as titanium oxide and zinc oxide, can achieve ILSS increases of about 60%–70% [34–39]. Notably, BP-1 particles deliver superior ILSS enhancement per unit specific gravity compared with many conventional fillers, making them particularly advantageous for aerospace and lunar applications where weight efficiency is critical. Their unique mineralogical and morphological features, including sharp edges and high surface roughness, likely promote enhanced interfacial adhesion and mechanical interlocking with the epoxy matrix, further elevating composite performance.

To further elucidate the influence of BP-1 lunar dust particles on the ILSS behavior of GFRPs, post-failure fracture surface analysis was conducted on ILSS specimens using SEM (Figure 4). For the neat epoxy composite (panels a–d), the side surface reveals extensive delamination across multiple layers with multiple delamination fronts, consistent with the strain localization patterns observed in the in-plane strain analysis of DIC. Interlayer fracture surfaces display pronounced river markings within the epoxy matrix (panel b), with magnified views showing brittle fracture features and poor fiber–matrix adhesion (panel c). Clean, exposed glass fibers (panel d) further confirm insufficient interfacial bonding in the unreinforced system.

In contrast, the composite containing 3 wt% BP-1 exhibits a fundamentally different fracture morphology (panels e–h).

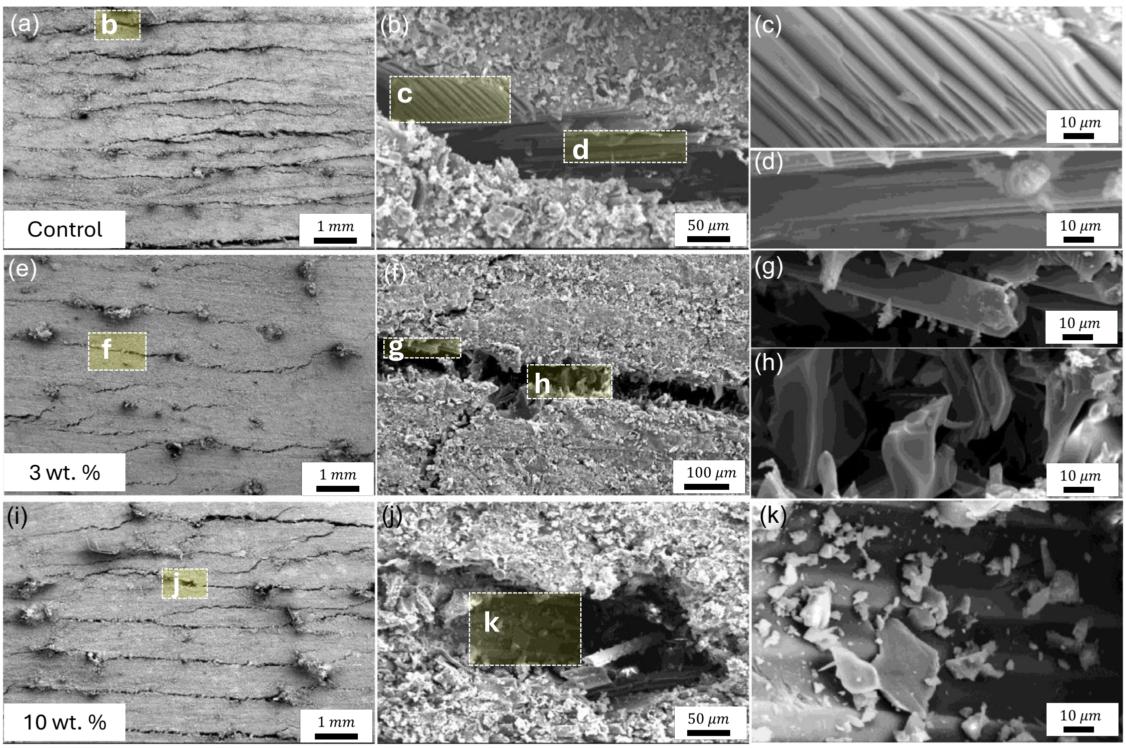

FIGURE 4 | (a) SEM image of the entire failure surface of the control samples, showing failure propagation throughout the sample. (b) Close-up view of region b in panel (a), highlighting the detailed failure features. (c) Close-up view of region c in panel (b), showing river marks in the epoxy matrix, indicative of brittle interfacial failure between the matrix and fibers. (d) Close-up view of region d in panel (b), revealing a clean surface of glass fibers, further confirming brittle fracture behavior. (e) SEM image of the entire failure surface of the 3 wt% BP-1 lunar dust-reinforced sample, showing failure propagation throughout the sample. (f) Close-up view of region f in panel (e), emphasizing the local deformation features. (g) Close-up view of region g in panel (f), where glass fibers are covered with sharp-edged BP-1 particles, indicating bridging between the epoxy matrix and fibers. (h) Close-up view of region h in panel (f), displaying shear bands and hackle marks, which are associated with enhanced ductility in the matrix due to improved interfacial bonding between the fibers and epoxy. (i) SEM image of the entire failure surface of the 10 wt% BP-1 lunar dust sample, showing failure throughout the sample. (j) Close-up view of region j in panel (i), focusing on the detailed surface features. (k) Close-up view of region k in panel (j), showing a smooth epoxy surface covering the fibers with an excess of BP-1 particles, a sign of poor adhesion and brittle fracture due to an overabundance of reinforcing particles.

Delamination is significantly reduced, and interlayer surfaces reveal sharp-edged BP-1 particles embedded at the fiber–matrix interface (panels f–g). These particles promote mechanical interlocking and act as local bridges between fibers and the surrounding matrix, thereby enhancing interfacial adhesion and load transfer. High-magnification images further reveal the presence of shear bands and hackle-like features within the matrix (panel h), which are characteristic of stable shear deformation and increased local ductility. These microstructural features directly underpin the more distributed shear strain fields observed by DIC and the enhanced ILSS measured in short-beam tests.

At higher concentrations (e.g., 10 wt%), however, the fracture mode reverts to a more brittle response (panels i–k). Widespread delamination again dominates, while fracture surfaces display smooth epoxy regions interspersed with dense BP-1 agglomerates. Excessive particle accumulation disrupts fiber–matrix continuity, suppresses effective particle bridging and crack-deflection mechanisms, and reintroduces localized stress concentrations, ultimately diminishing interfacial load transfer and damage tolerance.

Collectively, these observations demonstrate that BP-1 particles can significantly improve interfacial adhesion and fracture resistance in the composite laminate, but only within a controlled concentration range of 3%–7%. Moderate particle additions promote effective interlocking and crack deflection, whereas excessive filler content beyond this range undermines these benefits through agglomeration and weak interfacial bonding. For space-relevant applications, this highlights the importance of compositional optimization to ensure both mechanical reliability and long-term durability of regolith-reinforced composites.

2.3 | Low-Speed Impact Characteristics of Lunar Dust-Reinforced Composites

Impact resistance is a critical property for fiber-reinforced composites used in aerospace, automotive, and protective structures, where dynamic loading can trigger delamination, cracking, and premature failure. Enhancing the ability of composites to absorb and dissipate impact energy is therefore essential for durability and safety. One promising approach involves incorporating particulate fillers into the polymer matrix to reinforce the material and modify energy dissipation mechanisms. In this context, the present study evaluates the influence of BP-1 lunar dust particles on the impact resistance of GFRPs, measured by drop tower test, with results summarized in Figure 5.

Figure 5, panels (a) and (b) show normalized impact force–time and absorbed energy density–time responses from low-velocity impact tests on neat epoxy and BP-1-reinforced composites (3, 5, and 10 wt%). Normalization of force by panel thickness and impact energy by panel volume account for minor panel-to-panel thickness variations. The force–time curves exhibit the expected bell-shaped profile, with BP-1 addition increasing peak normalized force and prolonging its duration, which is evidence of improved energy absorption. Similarly, the absorbed energy density curves show an initial rise followed by a plateau, where the plateau reflects permanent inelastic deformation and internal damage such as delamination and matrix cracking.

At 3 wt% BP-1, composites exhibit a marked increase in absorbed energy density relative to neat epoxy laminates, demonstrating an enhanced ability to absorb and dissipate impact energy. As shown in panel (c), absorbed energy density continues to increase with filler content up to 10 wt%, while the absorbed-to-rebound energy ratio remains nearly unchanged. This indicates that BP-1 reinforcement enhances damage dissipation mechanisms without significantly altering the elastic recovery of the laminate. The improved impact performance likely arises from particle-induced crack deflection and local stress redistribution, which promote stable energy absorption under dynamic loading.

Panel (d) compares the low-velocity impact energy absorption capacity of BP-1–reinforced composites with that of composite systems containing conventional spherical, platelet, and needle-shaped fillers. Consistent with Figure 3d, the dataset was compiled from published studies and normalized to the neat-matrix baseline of each reference system. Conventional fillers show wide variability, with effects strongly dependent on particle morphology and surface chemistry. While some provide moderate improvements, others reduce impact resistance. A general trend suggests that fillers with higher specific gravity often achieve greater enhancements, albeit with substantial scatter. In contrast, BP-1 consistently delivers superior performance, exhibiting significant gains in absorbed energy per unit specific gravity while simultaneously suppressing detectable internal damage. This exceptional efficiency is attributed to the sharp-edged morphology and high surface area-to-volume ratio of BP-1 particles, which promote strong interfacial adhesion, effective load transfer, and efficient energy dissipation during impact events.

To further elucidate the mechanisms of improved damage tolerance of BP-1 reinforced composites, panel (e) summarizes the post-impact thermographic analysis in the form of heat maps at the mid-plane of the specimens. In the heat maps, dark spots indicate delamination damage; such features are prominent in the control samples but largely absent in BP-1–reinforced composites, even at the highest BP-1 concentrations tested. In BP-1-modified laminates, impact damage is confined primarily to shallow surface layers, reflecting improved interfacial bonding and greater structural integrity. Panel (f) schematically illustrates the post-impact damage morphologies obtained from visual inspection and thermographic measurements. As schematically illustrated, control specimens with neat epoxy exhibit extensive delamination beneath the impact site, whereas BP-1-reinforced composites show markedly reduced subsurface damage. Energy dissipation occurs predominantly through permanent inelastic deformation rather than widespread delamination, consistent with the enhanced shear plasticity observed in ILSS measurements. These findings highlight the role of BP-1 particles in suppressing delamination and promoting stable energy absorption under impact loading

These results demonstrate that incorporating BP-1 lunar dust simulant particles into GFRPs markedly improves impact resistance and damage tolerance. The enhanced absorbed energy density and reduced delamination reflect synergistic particle–matrix interactions, including improved interfacial adhesion and crack-bridging mechanisms. Notably, BP-1 outperforms many conventional fillers on a weight-normalized basis, underscoring its potential as a lightweight and sustainable reinforcement. Beyond aerospace composites, these findings provide compelling evidence for the viability of lunar regolith as an *in-situ* resource

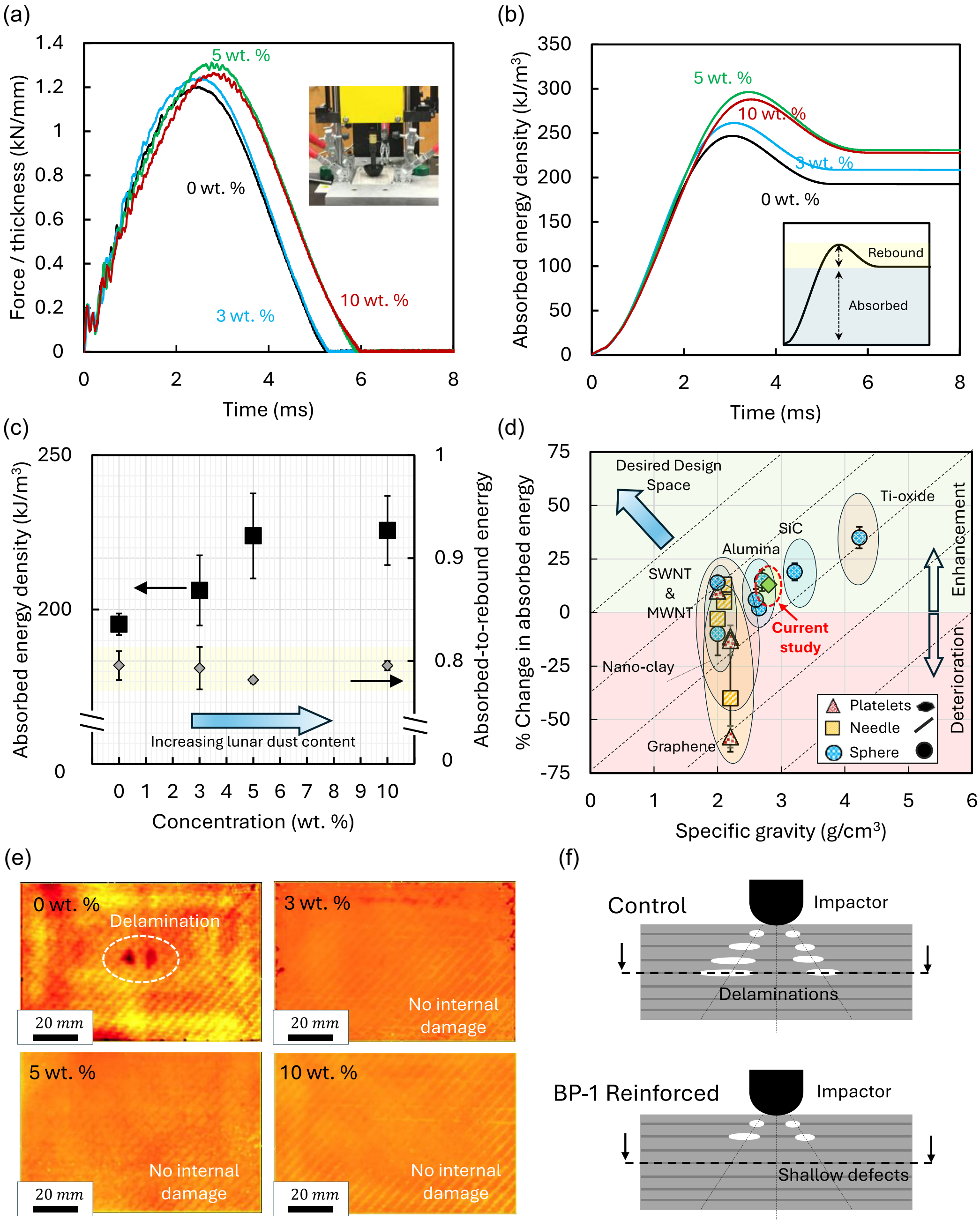

FIGURE 5 | (a) Low speed impact characteristics of GFRP panels with neat epoxy and BP-1 particle reinforced matrix at concentration of 3, 5, and 10 wt%. (a) Normalized impact force (Force/thickness) versus time curves and (b) absorbed energy density (Energy/Volume) versus time curves. (c) Variation of absorbed energy density and the absorbed-to-rebound energy density ratio as a function of BP-1 lunar dust simulant concentration, demonstrating a significant enhancement in absorbed energy density with minimal change in the absorbed-to-rebound energy density ratio as filler content increases. (d) Comparative chart showing the percentage improvement in low-speed impact energy absorption capacity achieved using BP-1 particles relative to common reinforcements, including spherical, platelet, and needle-shaped particles, as a function of filler-specific gravity. The chart highlights the superior performance of BP-1 particles compared to conventional fillers reported in the literature [21–25, 31–33, 22, 31, 40–42]. Data are normalized with respect to the neat epoxy baseline in each study; symbols denote mean relative changes, and error bars represent the range of reported values across different filler contents. (e) Post-damage flash thermographs heat maps sampled at the mid-plane of GFRP panels, highlighting impact-induced delamination beneath the impactor for the control group, while no visible damage was observed in BP-1 lunar dust-reinforced composites. (f) Schematic representation of observed damage patterns for the control and BP-1 lunar dust-reinforced samples.

for high-performance structural materials. Such reinforcements align with long-term goals of enabling durable, resource-efficient infrastructure for space exploration while simultaneously informing the design of advanced composites for terrestrial applications.

3 | Conclusions

We present a comprehensive investigation of the mechanical and fracture performance of glass fiber-reinforced composites enhanced with BP-1 lunar dust simulant particles, emphasizing their potential as sustainable reinforcement fillers for advanced composites. By systematically varying BP-1 concentration, we demonstrated significant improvements in ILSS, impact resistance, and energy dissipation, establishing BP-1 as a multifunctional additive for high-performance composites.

Our results show that modest BP-1 loadings strengthen interlaminar regions by impeding crack initiation and propagation, without substantially altering the intrinsic mechanical and fracture properties of the epoxy matrix. These enhancements are attributed to the angular morphology and high surface area of BP-1 particles, which promote strong interfacial adhesion, mechanical interlocking, and crack-bridging mechanisms.

In terms of impact performance, BP-1-reinforced composites exhibit greater energy absorption capacity and improved damage tolerance, as confirmed by normalized force–displacement curves, absorbed energy density measurements, and thermographic damage mapping. Even at higher concentrations, where many conventional fillers fail, BP-1 maintains structural integrity and damage tolerance, outperforming widely used reinforcement systems on a weight-normalized basis. Collectively, these findings position BP-1 lunar dust as a promising reinforcement for aerospace and protective composites where mechanical robustness and sustainability are paramount. Beyond Earth, they highlight the feasibility of leveraging lunar regolith as an *in-situ* resource to fabricate durable, lightweight structural components directly on the Moon.

The regolith contents examined in this study are intentionally limited to low weight fractions to evaluate reinforcement efficiency rather than material replacement. Even modest regolith additions yield substantial improvements in ILSS, impact resistance, and delamination suppression, making this approach particularly relevant for early-stage lunar missions where partial ISRU integration is more feasible than full *in-situ* manufacturing. While glass fibers were used here as a well-characterized baseline, the underlying matrix and interfacial toughening mechanisms are fiber-agnostic and directly applicable to future basalt fiber-based composite systems.

Future studies should extend this work to high-velocity impact, cyclic fatigue, and environmental aging to fully establish durability under operational conditions. Coupling experiments with microstructural modeling will enable optimized particle dispersion strategies, while extending BP-1 integration to hybrid and 3D-printed composites could expand applicability. A particularly compelling direction is the development of *in-situ* additive manufacturing processes capable of incorporating lunar dust into polymer matrices under vacuum and microgravity.

By advancing the understanding of extraterrestrial dust simulants as functional fillers, this study bridges materials design with sustainable space exploration strategies. The insights gained here establish a foundation for multifunctional, resource-efficient composites that meet the demands of both terrestrial engineering and future space infrastructure.

4 | Experimental Section

4.1 | Materials

The composites were fabricated using the hand layup technique, followed by vacuum bagging and curing at room temperature for 24 h. Saertex Biaxial Glass Fabric ($444\,g/m^2$), renowned for its excellent strength and flexibility, was carefully arranged in a mold as the reinforcement material. West Systems Resin 105, mixed with Hardener 206 in the manufacturer-recommended 5:1 ratio, served as the high-strength epoxy matrix. Each layer of the glass fabric was manually impregnated with the resin mixture to ensure complete wetting before being layered in a controlled sequence.

The epoxy system used in this study has a low room-temperature viscosity, approximately 600–900 cP prior to curing, which facilitates effective wetting of the fibers and dispersion of BP-1 regolith particles. This relatively low viscosity, combined with mechanical stirring, vacuum degassing, and ultrasonic treatment, enabled uniform particle dispersion at the regolith loadings investigated in this work. It is noted that the use of lower viscosity binder systems may further improve particle dispersion and enable higher regolith contents, which represents an important direction for future studies focused on maximizing ISRU.

Once the layup was complete, vacuum bag curing was performed. Vacuum was maintained throughout the curing process at a partial vacuum level of approximately −90 kPa (gauge) to ensure laminate consolidation and continuous removal of excess resin. Importantly, this process does not rely on resin volatilization; instead, surplus epoxy is mechanically removed via resin bleed into the breather/bleed fabric during vacuum bagging. No evidence of resin evaporation, incomplete curing, or void formation was observed in the cured laminates. The controlled bleed-off of excess epoxy may result in a slight increase in the effective regolith weight fraction within the cured matrix; however, this effect was consistent across samples and did not affect the qualitative trends or conclusions of the study. It is noted that sustained high-vacuum environments would require alternative low-volatility resin systems, which remains an important consideration for future *in-situ* lunar manufacturing. After curing, panels measuring 6 inches by 6 inches were fabricated and subsequently cut into short beam test specimens measuring 2 inches by 0.5 inches for mechanical property evaluation.

For the control samples, no filler material was added, and neat epoxy was employed. These control samples served as the baseline for comparison against the composites reinforced with lunar dust filler. For the lunar dust-reinforced samples, the process began by weighing the required amount of epoxy resin. A predetermined percentage of BP-1 lunar dust filler, based on the epoxy's weight, was then added to achieve specific filler concentrations of 1 and 10 wt%. The filler was carefully mixed into the resin to ensure uniform distribution and avoid clumping. To further enhance dispersion, the mixture was subjected to vacuum degassing for approximately 5 min, removing trapped air bubbles that could compromise the composite's mechanical properties. Following degassing, ultrasonic dispersion was carried out in a bath-type ultrasonic cleaner (water basin, power $\approx$ 100 W) for approximately 5 min to break down agglomerates and achieve finer, more uniform dispersion of the lunar dust particles within the epoxy matrix. The modified resin mixture was then applied to the glass fiber fabric using the same hand layup process as the control samples.

4.2 | Mechanical and Fracture Tests

4.2.1 | Tensile Test

Tensile tests were conducted on the epoxy systems to determine their mechanical properties, including ultimate tensile strength, Young's modulus, and elongation at break. The procedure followed ASTM International (ASTM) D638 [56], which specifies the standard method for testing tensile properties of plastics. Dog-bone-shaped specimens with Type IV geometry were fabricated with nominal dimensions as specified in ASTM D638. The samples were cured under standard conditions to ensure consistent properties. An electromechanical universal testing machine (Instron 5982) with a 50 kN load cell was used. Specimen edges were roughened and clamped using wedge grips to prevent slippage during loading. Tensile tests were carried out under displacement control with crosshead displacement rate of 5 mm/min, as per the ASTM standard. Load and displacement were recorded, and stress–strain curves were plotted. Strain was measured using a noncontact video extensometer (Instron SVE 2) attached to the test frame.

4.2.2 | Single-Edge Notched Bend Test

The fracture toughness of the epoxy matrix was evaluated using the SENB test, following ASTM D5045 [57] for linear-elastic fracture toughness of plastics. Rectangular specimens (50 × 10 × 5 mm) with a pre-machined notch were prepared. A sharp crack was introduced using a razor blade to ensure proper crack initiation. The tests were conducted on a three-point bending fixture attached to an electromechanical universal testing machine (Instron 5982) with a 50 kN load cell. The span-to-depth ratio was maintained at 4:1, as specified in ASTM D5045. Bending tests were carried out under displacement control with crosshead displacement rate of 10 mm/min. The critical stress intensity factor (K_{IC}) was calculated from the load–displacement data using ASTM-provided formulas. The fracture toughness was determined based on the maximum load and the specimen's geometry, providing insights into the epoxy matrix's resistance to crack propagation.

4.2.3 | Short-Beam Bending Test

The ILSS of the composite system was measured using the short-beam bend (SBB) test, following ASTM D2344 [58]. Composite laminate panels (150 × 100 × 6 mm) were fabricated, and SBB specimens with nominal dimensions of 40 × 12 × 6 mm were sectioned from these panels. The samples were polished to ensure flat surfaces and uniform thickness. The test was conducted on a three-point bending fixture with a span-to-thickness ratio of 4:1, as per ASTM D2344. The tests were conducted on a three-point bending fixture attached to an electromechanical universal testing machine (Instron 5982) with a 50 kN load cell. Bending tests were carried out under displacement control with crosshead displacement rate of 1 mm/min to ensure quasi-static conditions. ILSS was calculated from the maximum load using ASTM-prescribed formulas, offering insights into the role of fillers in enhancing interfacial shear properties in composite systems [59].

DIC was employed to capture full-field strain and displacement distributions during the SBB tests. The measurements were conducted using the Vic-2D DIC [55] system from Correlated Solutions. Specimen surfaces were coated with a high-contrast speckle pattern using black and white spray paint. This pattern ensured optimal correlation during DIC analysis. A high-resolution camera was used to capture images during the tests. The frame rate was adjusted based on the test's loading rate to ensure high temporal resolution. Images were acquired with a high-resolution camera (2448 × 2048 pixels) covering a field of view of approximately 40 × 34 mm, yielding a spatial scale of 16 μm/pixel. The average speckle size was 5–7 pixels (50–100 μm), and the correlation analysis was performed with a subset size of 21 pixels (∼300 μm mm) and a step size of 7 pixels (∼100 μm). Strain localization during the interlaminar shear deformation in the SBB tests were analyzed to correlate mechanical performance with observed fracture mechanisms.

4.3 | Low Speed Impact Test

Drop Testing Procedure and Methodology: The drop testing in this study was conducted using an instrumented Instron drop tower employing ASTM D7136/D7136M [60] standard for evaluating the damage resistance of fiber-reinforced PMCs under low-velocity impact conditions. The samples were subjected to a fixed impact energy of 20 J, ensuring consistency across all tests to facilitate comparative analysis of the effect of BP-1 lunar dust reinforcement.

Test Configuration: The experimental configuration included a fixture designed to replicate simply supported plate boundary conditions, as specified by the ASTM standard. The nominal dimensions of the samples were 150 × 100 × 5 mm, which are suitable for minimizing edge effects while maintaining structural relevance. A blunt hemispherical indenter with a diameter of 1 inch was utilized to deliver the impact load. This indenter geometry was chosen to induce a well-distributed contact stress and minimize local material puncture effects, thus emphasizing the bulk impact response.

Data Collection and Analysis: The instrumented drop tower provided real-time measurements of critical parameters such as impact force, impact energy, and impactor velocity throughout each test. The force–displacement and energy-time curves were extracted from the raw data for detailed analysis. The energy absorption characteristics were calculated using the procedure outlined in the ASTM standard, which involves integrating the area under the force–displacement curve. This approach ensured accurate quantification of the energy absorbed by the samples during impact, providing direct insight into the material's ability to dissipate kinetic energy and resist damage.

Rationale for Testing Parameters: The chosen impact energy level of 20 J represents a typical low-velocity impact scenario encountered in real-world applications, such as tool drops or minor collisions in aerospace and automotive components. The standardized boundary conditions and indenter geometry allowed for reproducibility and comparability with other studies in the literature. The nominal sample thickness of 5 mm ensured a balanced representation of the laminate architecture, including the effects of fiber orientation, matrix properties, and interfacial adhesion on the impact response.

4.4 | Scanning Electron Microscopy and Energy-Dispersive X-Ray Spectroscopy

The SEM and EDS analyses were performed to investigate the microstructural features of the composite samples and the elemental composition of the BP-1 lunar dust simulant particles. These techniques provided detailed insights into failure mechanisms, fiber–matrix interactions, and the distribution of BP-1 particles within the matrix.

The tested composite specimens were sectioned into small representative pieces and coated with a thin layer of gold using a Cressington 108 Auto Sputter Coater. This coating enhanced surface conductivity and prevented charging during SEM imaging, with the thickness carefully controlled to avoid obscuring fine microstructural details. SEM imaging was conducted using a Field Electron and Ion Company Quanta 450 field emission gun scanning electron microscope in high-vacuum mode, providing high-resolution images for topographical and morphological analysis.

EDS analysis, performed with an Oxford Instruments Aztec EDS system integrated into the SEM, was used to determine the elemental composition of key regions in the samples. This included the fibers, the epoxy matrix, and the BP-1 particles, enabling precise characterization of their chemical makeup and distribution.

The SEM images and EDS spectra were processed using proprietary software to extract relevant qualitative and quantitative data. High-resolution micrographs facilitated the evaluation of microstructural features, while EDS data offered compositional insights. The results were correlated with mechanical testing data to establish clear relationships between the microstructure, composition, and overall performance of the composite materials.

4.5 | Flash Thermography

The flash thermography tests were conducted at the Center for Non-Destructive Evaluation at Iowa State University to evaluate manufacturing-induced defects and impact-induced damage in composite panels before and after low-speed impact testing. The procedure followed standard guidelines for active thermographic inspection to ensure accuracy and reliability.

Before testing, the surfaces of the composite panels were coated with high-emissivity paint to ensure uniform thermal radiation during cooling. The paint used was Krylon Flat Black High Heat Paint, commonly applied in thermographic inspections due to its high emissivity and uniform application properties. Proper surface preparation and painting ensured that thermal energy emitted during cooling was not affected by variations in surface properties or reflectivity.

A high-intensity flashlamp system (Xenon 9101) was used to deliver a brief, uniform heat pulse to the surface of the panels. The flash energy was calibrated to ensure consistent heating across all samples while avoiding excessive localized heating that could cause material degradation. The panels were then monitored using a FLIR T1030sc thermal camera, which captured thermal images as the panels cooled. The camera was configured to record at a frame rate of 100 frames per second for a duration of 4 s, providing high temporal resolution for capturing transient thermal responses. The field of view was adjusted to cover the entire panel surface to detect defects and damage comprehensively.

The recorded thermographic images were analyzed to identify thermal anomalies indicative of defects. Before impact testing, all samples were examined to confirm the absence of manufacturing-induced defects. The pre-impact thermograms revealed minimal surface porosity and occasional fiber bunching, leaving small gaps in the surface layer. However, no significant structural defects were observed, ensuring that the test results after impact could be attributed solely to the damage induced by the low-speed impact.

Post-impact thermograms were then used to assess damage patterns, such as delamination, cracks, or resin-rich areas. Observations indicated that stitching in the fiberglass fabric was uniformly visible across all samples, confirming the consistency of the composite fabrication process.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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